Predicting Youth Marijuana Use

Hamda Hassan

Predicting Youth Marijuana Use

Decision Trees & Ensemble Models (Random Forest, Boosting)

- National Survey on Drug Use and Health (NSDUH)
- Focus: youth marijuana use behavior
- Goal: prediction + interpretation of key drivers

Research Question & Approach

How do demographic, socioeconomic, and social factors influence youth marijuana use, and how well can we predict it?

- **Approach**
- Binary classification (use vs non-use)
- Feature-based modeling (demographics, peers, family)
- Interpretable ML methods:
  - Decision Tree
  - Random Forest
  - Boosting

Data Overview

- **NSDUH Youth Dataset**
- ~10,000 youth respondents
- 79 selected variables (from larger survey)
- Key domains:
  - Demographics (sex, race, income)
  - Family environment (parental support/structure)
  - Peer influence (drug exposure among friends)
  - School & behavioral indicators

  Outcome Variable
- Marijuana use (binary: Yes / No)

Data Preparation & Feature Engineering

Data Preparation Steps

1. Selected theoretically relevant predictors
2. Created binary outcome (marijuana use: Yes/No)
3. Cleaned missing values (listwise deletion)
4. Encoded categorical variables for modeling

Key Predictor Groups

1. Peer influence (FRDMJMON, PRMJEVR2)
2. Family structure (PARHLPHW, IMOTHER, IFATHER)
3. Socioeconomic status (INCOME, POVERTY3)
4. Demographics (IRSEX, NEWRACE2)

Modeling Strategy & Train/Test Split

Modeling Setup

- 70% training / 30% testing split
- Same split used across all models
- Models used:
  - Decision Tree
  - Random Forest
  - Gradient Boosting (GBM)

Goal

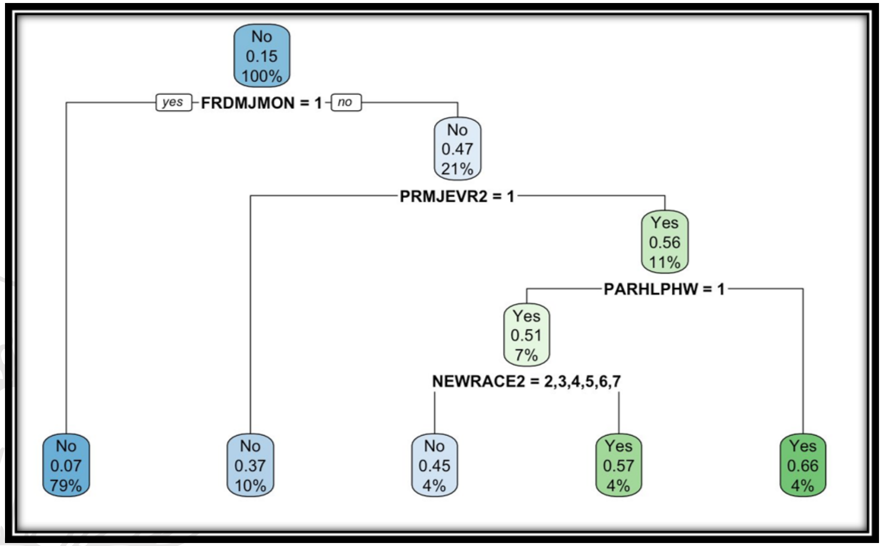
- Compare interpretability vs predictive performance

Decision Tree Results

- **Most Important Predictors**
- FRDMJMON (peer marijuana use) – strongest split
- PRMJEVR2 (peer exposure)
- PARHLPHW (parental support)
- NEWRACE2 (demographics)

Key Insight

- Peer influence drives first and strongest splits
- Family environment refines predictions
- Demographics appear deeper in the tree



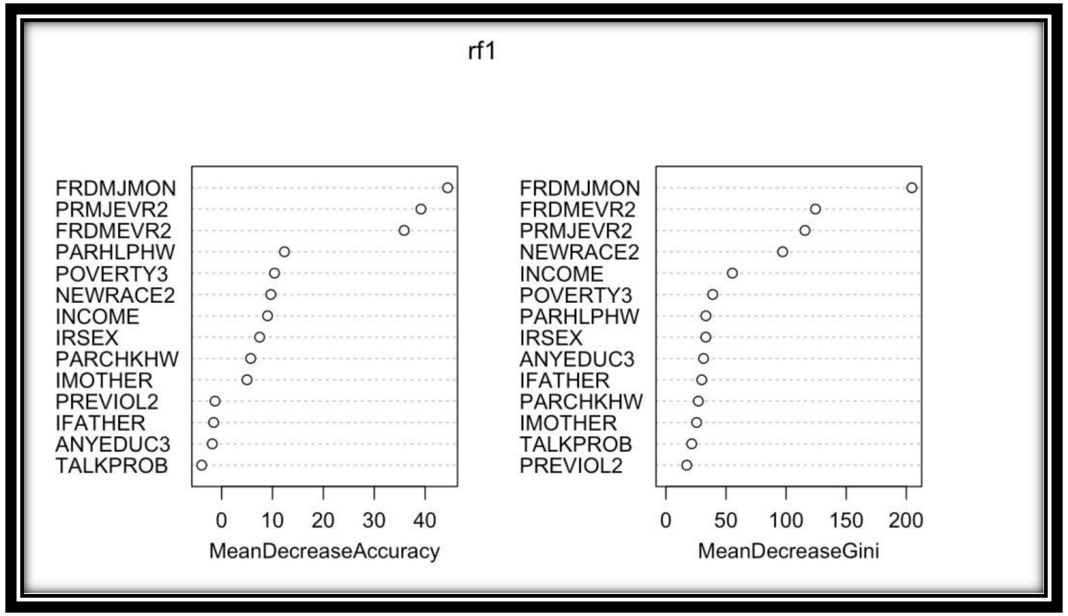
Random Forest Results

Top Variables

- FRDMJMON (dominant)
- PRMJEVR2
- FRDMEVR2
- INCOME
- POVERTY3

Key Insight

- Peer influence consistently strongest predictor
- Socioeconomic factors provide secondary signal
- More stable than single tree



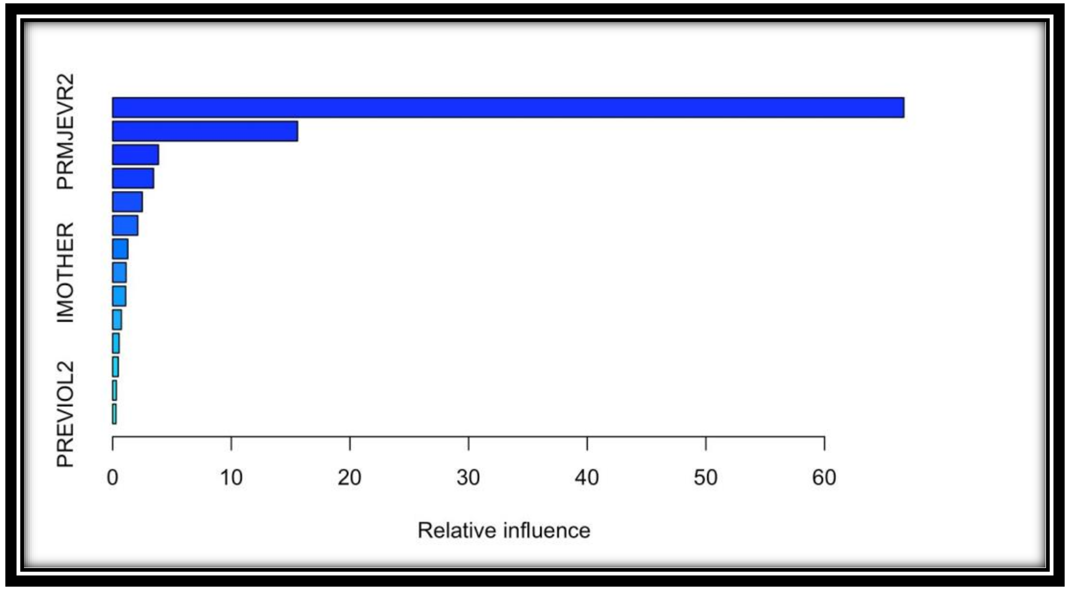
Boosting Results

Variable Importance

- FRDMJMON: 66.7%
- PRMJEVR2: 15.6%
- FRDMEVR2: 3.9%
- Others: minimal contribution

Key Insight

- Model heavily concentrates on peer influence
- Iteratively prioritizes strongest predictors



Model Performance (Confusion Matrix)

Performance Summary

- Accuracy: 85.8%
- Strong prediction of non-use
- Weak detection of users

Class Imbalance

- ~85% non-users
- ~15% users

Interpretation Across Data Types

Different Representations of Marijuana Use

- Binary (Yes/No)
- Continuous (days used per year)
- Categorical (usage intensity)

Key Insight

- Binary = best for classification
- Continuous = captures intensity
- Categorical = more interpretable but less precise

Challenges & Data Leakage Fix

Challenges

- Initial data leakage issue
- Outcome-related variables included as predictors
- Required restructuring of datasets

Fix

- Separated datasets by model type
- Ensured no overlap between predictors and outcomes

Conclusion & Ethical Considerations

Key Findings

- Peer influence is the strongest predictor
- Family environment is secondary
- Socioeconomic factors are less influential
- Models perform better on non-use than use

Ethical Note

- Results show correlation, not causation
- Findings should not be used to stigmatize individuals

Thank you!

Hamda Hassan